

Clockless Alignment: Developing an Algorithm to Infer Neural–Behavior Correspondence

1st Cycle Integrated Project in Engineering Physics

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1 Introduction and Project Scope

A recurring practical problem in systems neuroscience is that behavioural data and neural recordings are usually collected on physically separate computers, each running its own internal clock. To relate a behavioural event (for instance, the moment an animal pulls a lever to receive a water reward) to what the brain was doing at that exact instant, the two independent time bases must first be related to one another — a process generally called temporal alignment [1], or synchronisation. The conventional solution is a hardware synchronisation signal: a digital pulse sent simultaneously to both recording systems, giving a small number of timestamps that are known, by construction, to refer to the same physical instant on both clocks. Fitting a line through these shared timestamps yields a clock-rate (stretch) and offset correction that can then be applied to every other event.

This project addresses the situation where that hardware signal is missing or was not correctly recorded for a subset of sessions, leaving only the behavioural log and the raw neural recording, with no a priori shared reference points between them. The work is organised in three phases, each motivated by the partial failure of the one before it:

- **Phase 0 (Video alignment).** Before tackling the harder ephys-clock problem, the same underlying idea — detect a physical correlate of the behavioural event in an independently-clocked recording, and fit a clock warp from matched timestamps — was first developed and tested on simultaneous video recordings, where there is a clear, visually inspectable physical correlate of the event (droplet). This established the matching and warp-fitting machinery reused for the ephys problem, and confirmed it could reach second-scale alignment precision.
- **Phase 1 (Continuous-feature ephys alignment).** For the sessions with no functional hardware sync, several hundred continuous signals derived from the raw electrophysiology (artefact detectors, multi-unit activity envelopes, local field potential power in different frequency bands, and others) were engineered and tested as candidate event detectors, by direct analogy with the video approach. None proved specific enough to the behavioural event to support precise alignment.
- **Phase 2 (Spike-sorted anchor alignment).** Rather than a population-level continuous signal, a single, individually-isolated neuron with a sharp and reliable response to the behavioural event was used directly as a biological clock. This phase achieved a large improvement in alignment precision, but still has limitations addressed in the future directions.

2 Phase 0: Video–Behaviour Alignment

2.1 Motivation and Method

The two recording systems — a behaviour-logging computer and a video camera — run on independent clocks that drift apart over the course of a session.

Detection was framed as a confidence-weighted ensemble vote rather than a simple threshold on one signal. Two main complementary cues were combined for each candidate video frame within a short window after a logged behavioural event: (i) a principal-component projection of pixel intensity in a small region of interest (where the droplet is predicted to be located) and (ii) a normalised cross-correlation (template-matching) score against a reference image crop of the droplet. The highest-scoring candidate above a minimum-confidence threshold was accepted as the matched event for that trial.

A grid search over the relative weighting of the principal-component cue and the minimum-confidence threshold (fifteen configurations in total) was used to select the best-performing combination, since the relative reliability of the two main cues could not be predicted analytically and needed to be determined empirically. The best configuration achieved a mean absolute error of 3.09 s and root-mean-square error of 3.68 s on held-out trials — that is, on the subset of matched trials excluded from the fitting step itself and used only to check how well the resulting warp generalises to data it had no part in determining — with all 138 trials in the test session successfully matched (100%), and a fitted clock relationship $t^{\text{video}} = 1.0021 t^{\text{bhv}} + 1.15 \text{ s}$, corresponding to a total accumulated clock drift of $\sim 4.8 \text{ s}$ over the $\sim 2300 \text{ s}$ session. The median confidence across matched trials was 0.8.

Known limitation, carried forward. The optimiser used to fit the clock warp was constrained to search for an offset within a fixed bound (initially $\pm 5 \text{ s}$); this bound was reached during fitting, meaning the true offset may have been outside the search range and the optimiser returned the best value it was permitted to find rather than the true best value. This produced a substantially inflated raw error figure in the initial sweep output, later corrected by widening the search bound to $\pm 100 \text{ s}$ for later sessions.

2.2 Diagnostics

Figure 1 shows the full diagnostic picture for the best configuration. The top panel plots behaviour time against video time for all 138 matched trials, coloured by detection confidence; the points track the fitted line essentially perfectly across the full session range, with no visible curvature, confirming that a single linear (constant clock-rate) model is an adequate description of the drift for the whole session. The middle panel re-expresses the same fit as a clock offset per trial, making the drift trend directly visible as a near-perfectly linear climb across the session — exactly the signature expected from a small, constant clock-rate mismatch between the two systems, rather than an erratic or trial-dependent error. The bottom panel checks whether detection confidence predicts alignment error: a weak negative trend shows that trials with

larger accumulated clock offsets are mildly associated with slightly lower confidence, consistent with matching becoming marginally harder later in the session as the offset grows, though the effect is small.

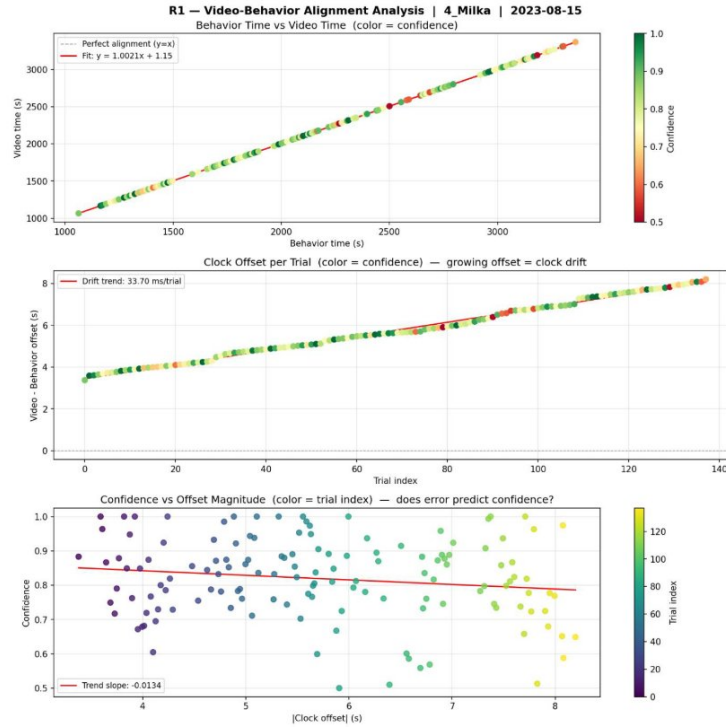


Fig. 1: Video-behaviour alignment diagnostics for a representative session. Top: Behaviour time vs. video time for 138 matched trials, coloured by detection confidence; red line is the fitted warp. Middle: Clock offset per trial, showing the linear drift trend directly. Bottom: Confidence vs. |offset| magnitude, coloured by trial index; weak negative trend.

A separate diagnostic, independent of the fitted warp itself, is the raw first-principal-component trace overlaid with the logged event times (Figure 2). The full-session trace (top panel) shows a clear qualitative regime change partway through the session: before this point the trace sits in a low, tightly-bounded quiescent band, consistent with the animal not yet engaged in the task; afterwards the trace becomes both higher in mean and substantially noisier, and this is exactly where the great majority of logged events are concentrated. The zoomed view (bottom panel) resolves this further: individual event times align with local structure in the trace, though the trace is visibly noisy at the single-event level, which is precisely why the confidence-weighted ensemble of multiple cues was used for detection rather than thresholding this one signal alone.

3 Phase 1: Continuous-Feature Ephys Alignment

3.1 Problem and Approach

In the sessions of primary interest, the dedicated hardware synchronisation channel — a digital line recorded alongside the neural data that pulses every time the behaviour computer

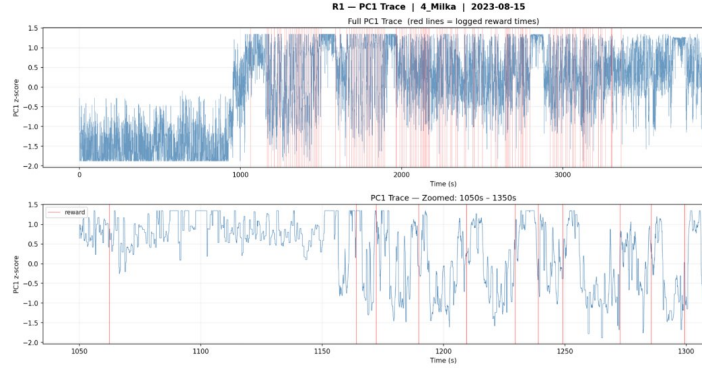


Fig. 2: First-principal-component trace vs. logged event times for a representative session. Top: Full-session trace (blue) with event times overlaid (red vertical lines). Bottom: Zoomed view showing individual event times against the trace.

marks an event — was confirmed to be flat throughout the recording, meaning no synchronisation pulses had actually been captured, despite the recording hardware being present and configured to record this channel. With no hardware ground truth available for these sessions, the problem reduces to the same one solved conceptually in Phase 0: find some signal in the electrophysiology recording that reliably and specifically marks the moment of the behavioural event, then use matched event pairs to fit a clock warp exactly as before.

By direct analogy with the video approach, several candidate continuous signals were engineered from the raw recording: a hypothesised electromagnetic artefact from the reward-delivery mechanism itself that would show up as a common mode across all channels; the envelope of high-frequency multi-unit spiking activity; and the power of the local field potential in several different frequency bands, computed separately across depth along the probe and condensed via principal component analysis. When these initial signals proved insufficient, a substantially larger set of features (several hundred in total) was engineered, including frequency-band-specific power per probe depth segment, a supervised reward-modulation index, a common-average-referenced version of the multi-unit signal (intended to remove shared artefact across channels before further processing), a more noise-robust spike-density estimate, and a measure of coupling between the phase of slow oscillations and the amplitude of fast oscillations.

3.2 Result: Negative

No feature achieved usable specificity for alignment. Detection performance was quantified using the standard precision/recall-based F1 score, comparing detected candidate peaks in each feature against the logged behavioural events; the highest F1 scores across all features tested were modest (Table 1), well below what would be needed for reliable per-trial alignment. The structural problem common to every feature: each one responds to a broad mixture of general movement, in addition to (or instead of) the specific behavioural event of interest, so the number of candidate peaks vastly outnumbers the true number of events — one representative feature, for instance, produced over ten thousand candidate peaks against only a few hundred true

events in a single session (Fig. 3). With a detection window of a few hundred milliseconds, a substantial fraction of peaks will coincidentally fall near a true event purely by chance even if the feature carries no real information about event timing at all; low F1 therefore does not necessarily mean a feature contains no real signal, only that on its own it is not specific enough to serve as a reliable clock anchor.

Feature type	F1	Hit rate	Med. error (ms)
Phase-amplitude coupling (best)	0.118	0.318	217
Phase-amplitude coupling (2nd)	0.114	0.298	170
Reward-modulation index (best)	0.094	0.500	165
Reward-modulation index (2nd)	0.093	0.530	194
Reward-modulation index (3rd)	0.091	0.542	189

Tab. 1: Best-performing continuous features by F1 score, representative session.

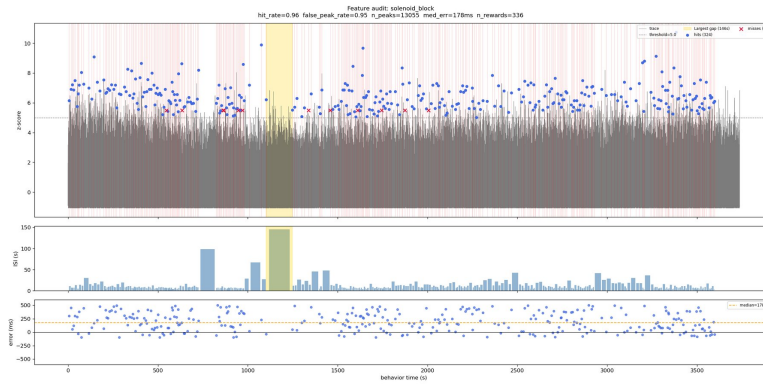


Fig. 3: Feature audit for a depth-blocked artefact-detection feature, full session. A high nominal hit rate is accompanied by an equally high false-peak rate and a large number of total candidate peaks relative to true events, confirming the feature is sensitive but not specific to the behavioural event.

A two-stage detection strategy was also tried, in which a primary artefact-detector candidate anchored each trial and a secondary, more locally-specific signal was allowed to refine the estimated time within a narrow window around that anchor. This did not resolve the underlying specificity problem for the broader feature set, since the artefact assumption didn’t hold as expected.

Trial-aligned response plots computed using the resulting alignment (root-mean-square error around 167 ms) were flat for nearly all features tested (Fig. 4 shows the one partial exception, with visible but small trial-locked structure).

Method	RMSE (ms)	Notes
Best single continuous feature	144	Still unusable
Continuous features fused	179	No improvement
Single-unit anchor (Phase 2)	10.75	15-fold improvement

Tab. 2: Alignment quality, best continuous methods vs. single-unit anchor, representative session.

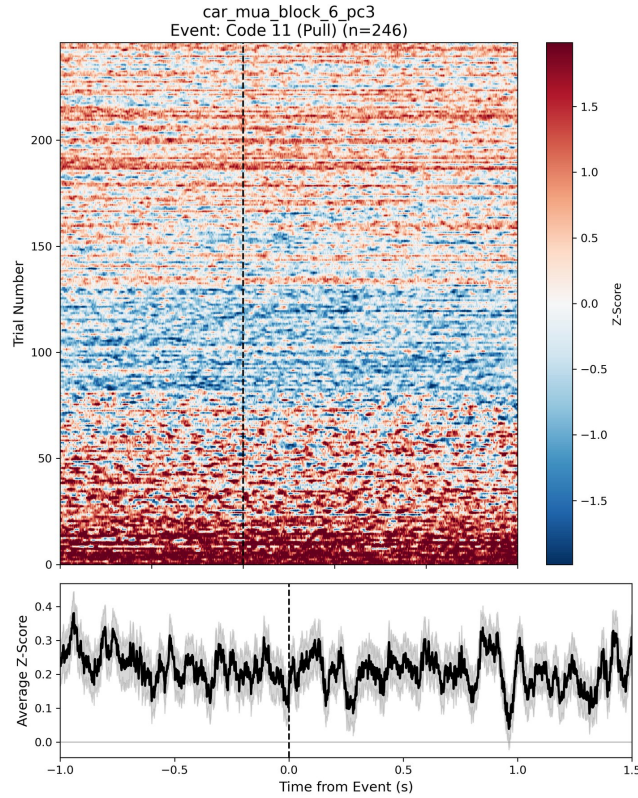


Fig. 4: Trial-resolved response for one multi-unit-activity-derived feature, aligned to a behavioural event ($n = 246$ trials) — the closest any continuous feature came to visible trial-locked structure, still too weak relative to trial-to-trial variability to serve as a precise alignment anchor.

3.3 Why a Different Approach Was Needed

The failure of feature-based alignment motivated a fundamentally different strategy: rather than searching for a population-level continuous signal that merely correlates with the behavioural event, search instead for an individual, well-isolated neuron that fires reliably at a fixed, short latency after the event.

A unit-selection procedure was developed that scores every well-isolated unit in a recording by the contrast between its peak firing rate shortly after the behavioural event and its baseline firing rate beforehand, with an additional requirement that the unit fire at least once within the expected response window on a sufficiently high fraction of trials.

3.4 Computational Logistics

A substantial fraction of the total time spent on this project was not algorithmic but logistical, and is worth documenting explicitly, since it shaped which approaches were practical to even attempt. A single raw electrophysiology recording for one session is large by ordinary standards: the high-frequency band of one probe recording for roughly an hour of data, sampled at 30 kHz across several hundred channels simultaneously, occupies on the order of 70–80 GB on disk, with a lower-frequency companion file occupying a further few gigabytes. With eight such probe recordings across the dataset, and several hundred candidate features computed per recording in Phase 1, working with this data required batching.

This had consequences for how the pipeline was engineered. First, every raw file is read exactly once, in fixed-size, non-overlapping or lightly overlapping chunks (on the order of one minute of recording at a time), with every derived feature computed during that single pass and written to a compact intermediate cache file used for all later processing. This meant the expensive read-and-extract step only had to happen once per session; the bottleneck that remained was not re-reading the raw data, but the sheer number of distinct features being experimented with on each pass.

4 Phase 2: Spike-Sorted Anchor Alignment and Neural Circuit Analysis

4.1 Data and Method

Eight probe recordings, spanning two animals and multiple sessions, were analysed (Table 3), all using the same recording technology (384-channel silicon probes sampled at 30 kHz) and the same automated spike-sorting pipeline, with manual curation to label well-isolated single units.

The clock-warp procedure for a chosen anchor unit works as follows. First, the behavioural log is zero-anchored to a fixed reference event marking the start of the recording, so both logs share a common time origin. Then, for each behavioural event, a predicted spike time is computed by adding the anchor unit’s previously-measured peak response latency to the event’s behavioural timestamp; the nearest actual spike from that unit within a narrow window around this prediction is taken as the matched ephys time for that trial. Trials with no spike in the window are simply left unmatched. Once every trial has been processed this way, an iterative outlier-rejection step removes any matched pairs whose residual from a provisional linear fit exceeds three standard deviations, the fit is recomputed, and the process repeats a small fixed number of times; the final linear fit (rate and offset) on the retained pairs is the clock warp used for that session. For one session three different units were instead combined: each contributes its own per-trial estimate, and a weighted median across the contributing units’ estimates is taken per trial. This improved the per-trial precision substantially over the best available single unit in that particular session.

Session	Probe	Anchor latency	Pairs	RMSE (ms)
Animal A, session 1	probe 0	215 ms	276	10.75
Animal A, session 1	probe 1	305 ms	308	18.42
Animal A, session 4	probe 0	175 ms	115	13.57
Animal A, session 4	probe 1	175 ms	123	18.66
Animal B, session 1	probe 1	25 ms	127	16.44
Animal B, session 4	probe 0	55 ms	210	6.12
Animal B, session 4	probe 1	-95 ms	211	8.77
Animal B, session 2	probe 0	3-unit fusion	140	1.62

Tab. 3: Sessions and alignment statistics across the dataset.

4.2 Independent Validation Against Hardware Sync

The non-corrupt sessions provided an opportunity for an independent check: the hardware-derived clock warp (computed entirely independently of any spike data) was compared directly against the single-unit-anchor warp computed for the same session. The two warps agreed to within 6.98 ms root-mean-square error on predicted event time across several hundred matched pairs. This level of agreement is well within the tolerance used elsewhere in this work, and is a meaningfully stronger form of evidence than the internal self-consistency of the matching-and-fitting procedure alone, since it compares the method’s output against a ground truth.

4.3 Alignment Diagnostics

Figure 5 shows alignment diagnostics for the primary test session (276 matched pairs, $\text{RMSE} = 10.75 \text{ ms}$): the fitted clock-rate is constant over the session. Figure 6 shows the anchor unit itself, firing reliably around 215 ms after the behavioural event on the large majority of trials.

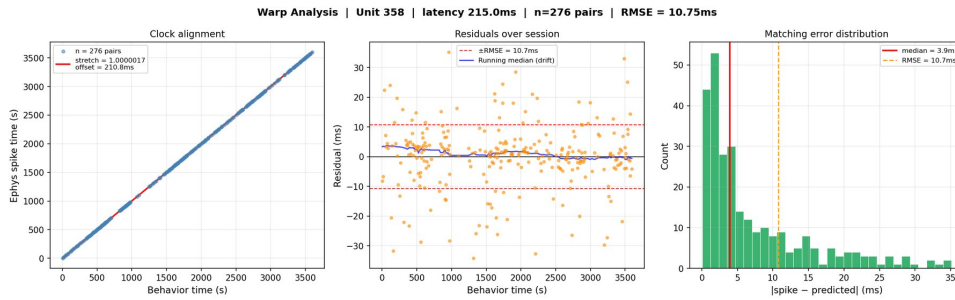


Fig. 5: Warp diagnostic, primary test session. Left: Behaviour vs. ephys time, 276 matched pairs; fitted warp. Centre: Residuals over session, flat running median confirms no nonlinear drift. Right: Matching error distribution; median 3.9 ms, RMSE 10.75 ms.

4.4 Neural Circuit Analysis

Units were classified by where their largest deviation from baseline occurred: a unit with its strongest response shortly before the event was labelled motor-related; one with its strongest response shortly after was labelled reward-related; a unit whose firing rate dropped well below baseline after the event was labelled suppressed; and any unit with no response exceeding a fixed statistical threshold in either direction was labelled unresponsive.

Figure 8 shows the timing structure across the whole recorded population, and Figure 10 sorts every unit in the recording by its peak response latency.

5 Further directions and conclusion

Despite the favourable error figures reported above, a closer methodological review of the single-unit-anchor approach surfaced a limitation that the aggregate error metric alone does not

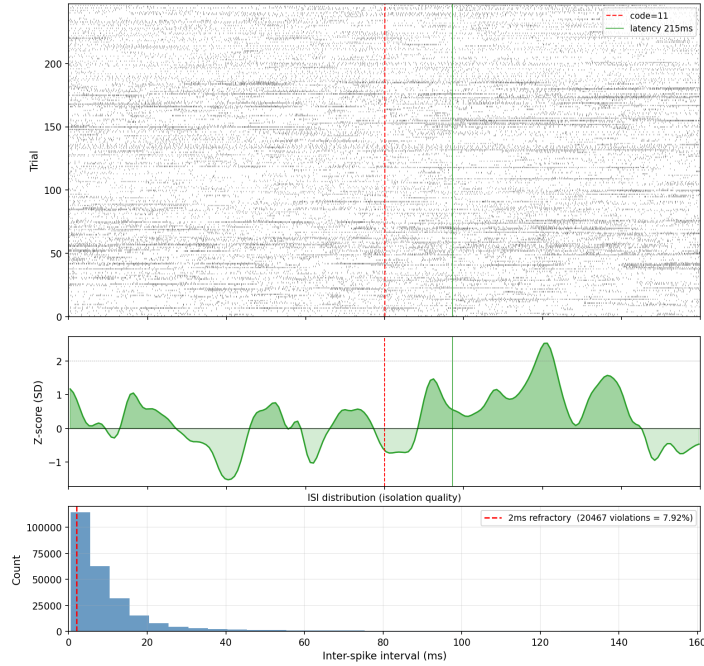


Fig. 6: Anchor unit, primary test session. Top: Raster across 246 trials, green line: response latency window. Middle: Trial-averaged response, baseline-normalised. Bottom: Inter-spike interval distribution, showing the elevated short-interval rate.

expose. The matching procedure scores and selects a unit by the contrast between its peak and baseline firing rate, and then matches each behavioural event to the *single nearest spike* from that unit within a narrow window around the unit’s expected response latency. For a genuinely bursty anchor unit, however, this window frequently does not contain one isolated, unambiguous spike — it contains a small cluster of several spikes fired in quick succession around the event. The nearest-spike rule selects one member of that cluster as if it were uniquely the event-locked spike, but this choice is, in an important sense, arbitrary: any other spike in the same local cluster is, a priori, an almost equally plausible candidate, and choosing a different one would shift that trial’s matched timestamp pair, and therefore the fitted clock warp itself, by however far apart the cluster’s spikes happen to be in time.

This is a structural ambiguity in the matching procedure, not ordinary measurement noise that would be expected to average out with more trials. The iterative outlier-rejection step only removes matches that are clear outliers *relative to the already-fitted warp*; it provides no independent evidence that the matches it retains are the temporally correct ones, only that they are mutually consistent with one another and with the fit they themselves helped produce. Likewise, the independent validation against a hardware reference signal (Sec. 4.2) confirms that the method’s aggregate output agrees with ground truth in the specific sessions where such ground truth happens to be available, but it does not by itself establish that the underlying single-spike selection logic is correct on a trial-by-trial basis.

A population-level alternative is the next step, informed by independent recordings from other sessions with genuine ground truth available. These additional sessions indicate that

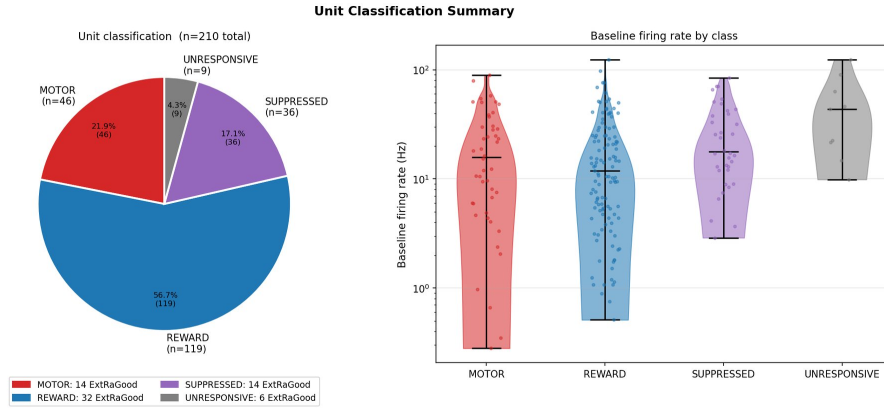


Fig. 7: Unit classification, primary test session ($n = 210$). Left: Class proportions. Right: Baseline firing rate by class (log scale); motor-related and suppressed units have higher baseline rates than reward-related units.

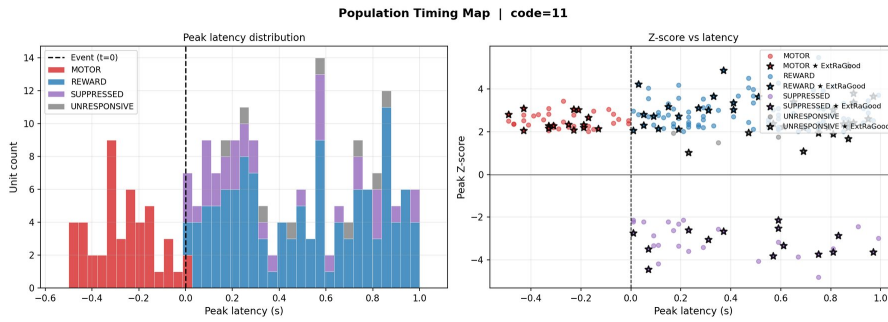


Fig. 8: Population timing map, primary test session. Left: Peak response-latency histogram by class. Right: Peak response magnitude vs. latency.

the units most reliably time-locked to the behavioural event are consistently of a particular physiological type: comparatively low-baseline, bursty units. Their individual firing patterns could in principle be averaged together into a single representative template. This template would necessarily be broader in time than any one contributing unit's individual burst, but the averaging process would still preserve a genuine peak at the event time, since the underlying time-locking property is shared across the contributing units, while any activity not actually locked to the event would simply average out toward a flat baseline rather than contributing a false peak. This template could then be cross-correlated against the full session's population spiking activity as a single combined signal, with a high match threshold chosen so that the resulting number of detected candidate events is comparable to the true number of behavioural events. This would directly target the same specificity problem that Phase 1 could not solve, but using spike-level rather than population-averaged continuous-signal information, and would remove the single-spike arbitrariness identified above by basing each detection on a multi-unit, multi-spike pattern match rather than an arbitrary single-spike selection rule.

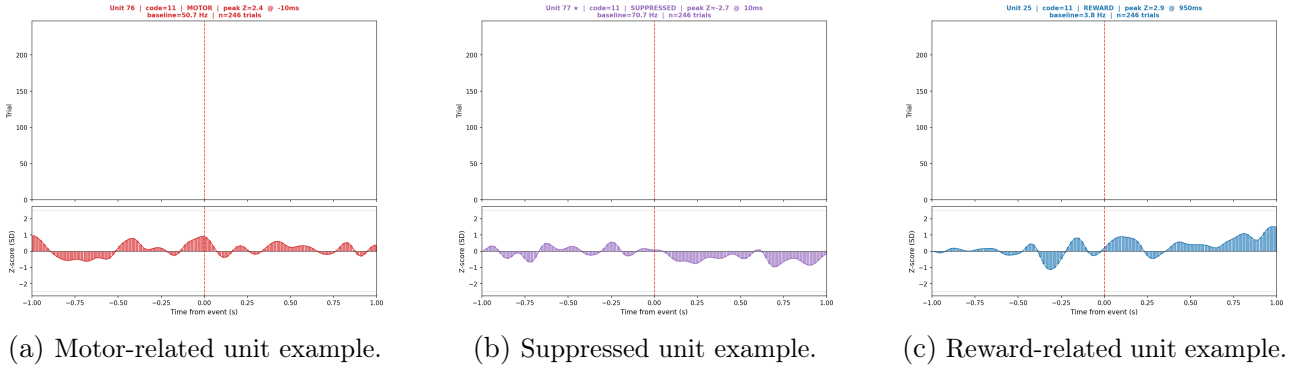


Fig. 9: Representative units from each response class, primary test session. Each panel: raster (top) and trial-averaged response (bottom).

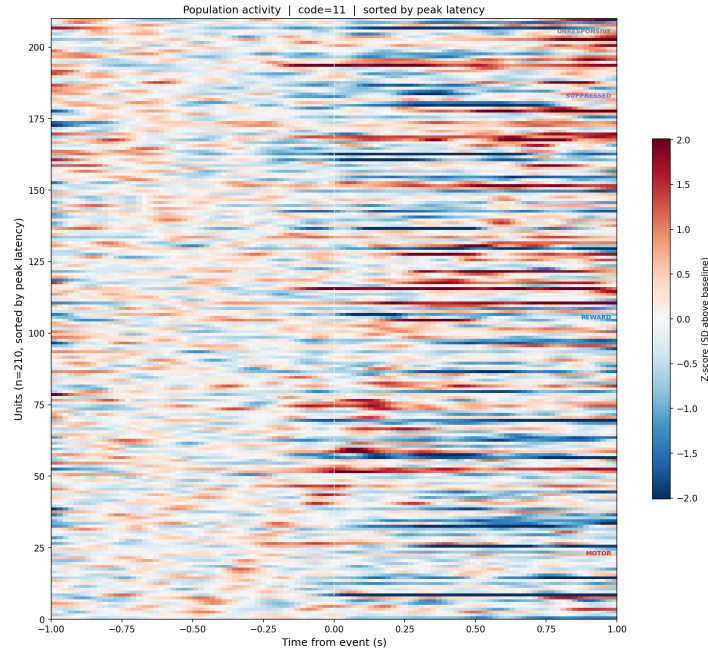


Fig. 10: Population activity heatmap, primary test session ($n = 210$). Units sorted by peak response latency; colour = deviation from baseline. The temporal cascade from pre-event to post-event activity is visible as a diagonal banding pattern.

References

- [1] Alex H Williams et al. “Discovering Precise Temporal Patterns in Large-Scale Neural Recordings”. In: *Neuron* 105.2 (2020), pp. 246–259.